

## 8.2 Classwork: Rational Functions — Toward the Maturità

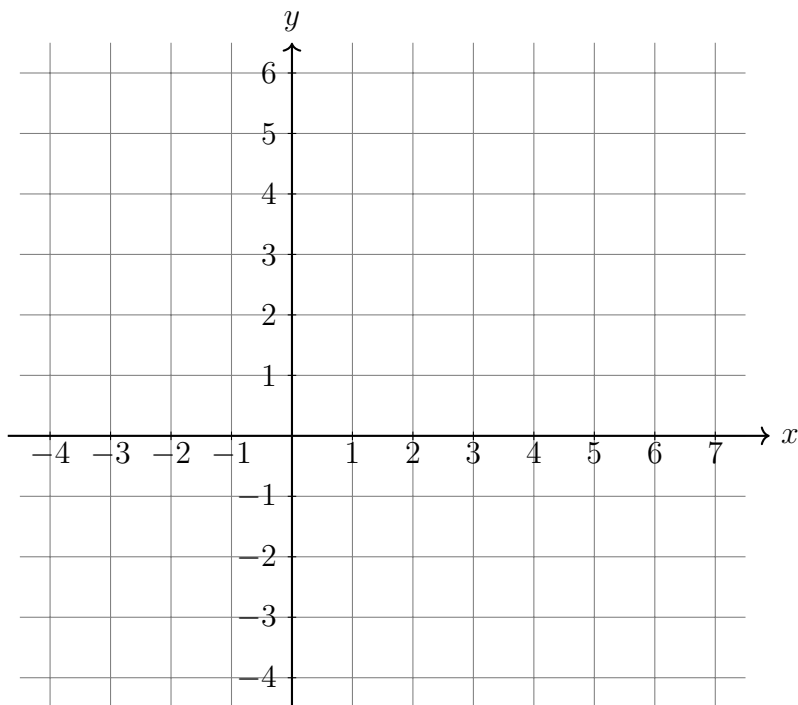
1. Consider  $f(x) = \frac{x+3}{x-2}$ .

(a) State the equation of the vertical asymptote.

(b) Verify by long division (or otherwise) that  $f(x) = 1 + \frac{5}{x-2}$ . Use this form to state the equation of the horizontal asymptote.

(c) Find the  $x$ -intercept and the  $y$ -intercept.

(d) Sketch  $f$  on the axes below. Mark and label the asymptotes and intercepts.



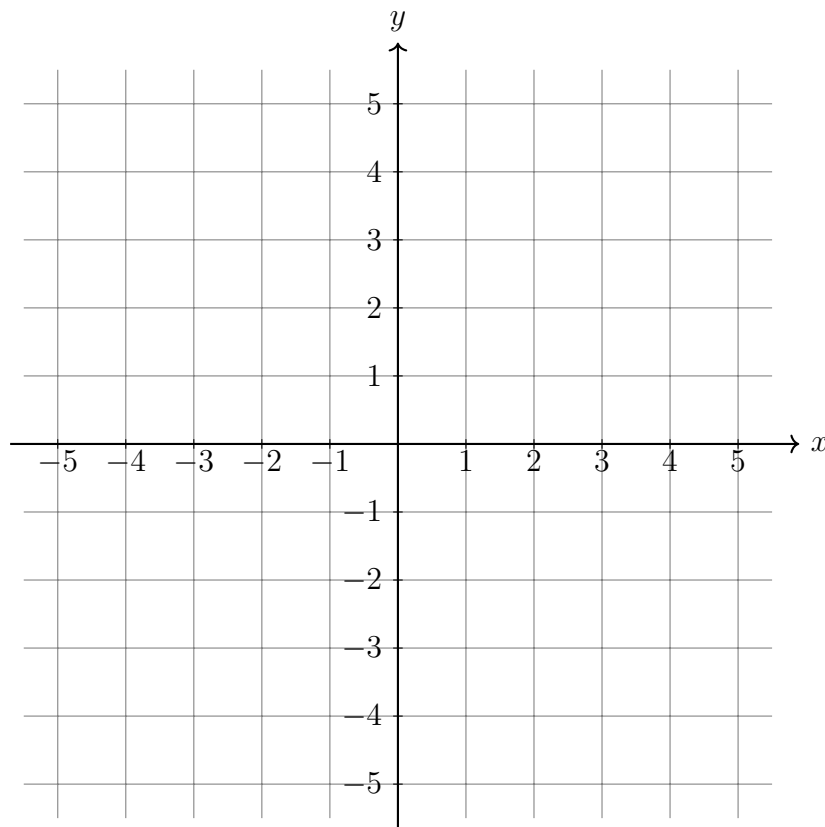
2. Consider the family of functions  $f_a(x) = \frac{x-2}{x-a}$ , with  $a \neq 2$ .

(a) For an arbitrary value of  $a$ , state the equation of the vertical asymptote of  $f_a$ .

(b) Find the horizontal asymptote of  $f_a$ . Does its equation depend on  $a$ ?

(c) Compute  $f_a(2)$ . What does this tell you about a point shared by every curve in the family?

(d) Now set  $a = -1$ . Find the  $y$ -intercept of  $f_{-1}$  and sketch its graph on the axes below. Mark the asymptotes (dashed) and the intercepts.



3. Consider  $g(x) = \frac{x^2 + x}{x^2 + 1}$ .

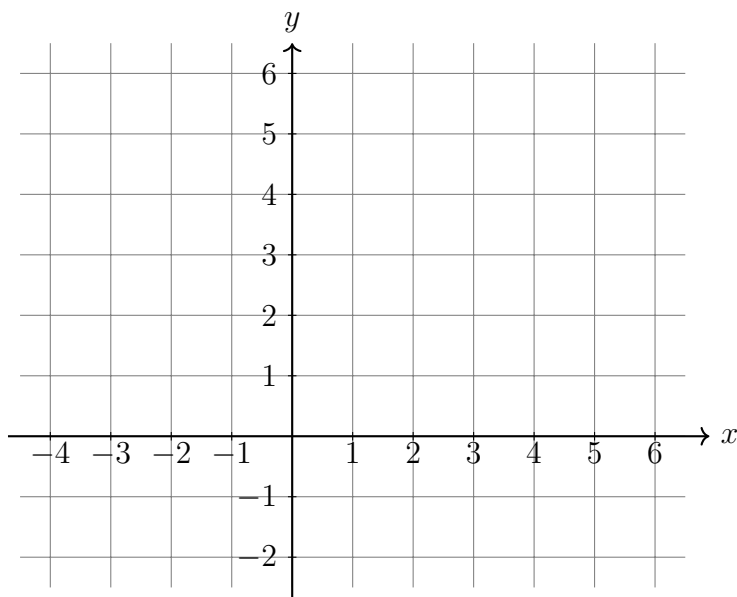
(a) State the domain of  $g$ . (Hint: look at the denominator carefully — can it equal zero?)

(b) Compute  $\lim_{x \rightarrow +\infty} g(x)$  and  $\lim_{x \rightarrow -\infty} g(x)$ . Hence state the horizontal asymptote.

(c) Find the  $x$ -intercepts (note: there are two) and the  $y$ -intercept.

(d) Complete the table of values, then sketch the graph below.

| $x$    | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 5 |
|--------|----|----|----|---|---|---|---|---|
| $g(x)$ |    |    |    |   |   |   |   |   |



4. Consider the family of functions  $f_a(x) = \frac{x^2 - ax}{x^2 - a}$ , with  $a \neq 0$ .

The function  $g$  from Problem 3 is the member  $f_{-1}$  (check: substituting  $a = -1$  gives  $g(x) = (x^2 + x)/(x^2 + 1)$ ).

(a) Find the domain of  $f_a$  in each of the following cases.

(i)  $a < 0$  (for example  $a = -1$ ,  $a = -3$ ):

(ii)  $a > 0$  (for example  $a = 4$ ):

(b) State the equation of the horizontal asymptote of  $f_a$ . (Hint: same idea as Problem 3(b) — compare leading terms.)

(c) Set  $f_a(x)$  equal to the horizontal asymptote you found in (b), and solve for  $x$ . Show that the solution does not depend on  $a$  (provided  $a \neq 1$ ). Conclude that every curve  $\Omega_a$  in the family passes through one common point on its horizontal asymptote. State this point.

(d) Compute  $f_a(0)$ . Then show, using the quotient rule, that  $f'_a(0) = 1$  for every  $a \neq 0$ . Hence write down the equation of the line tangent to  $\Omega_a$  at the origin. Is this line the same for every  $a$ ?

5. Consider  $h(x) = \frac{x^3 + 4}{x^2}$ .

(a) State the domain of  $h$ , and the equation of the vertical asymptote.

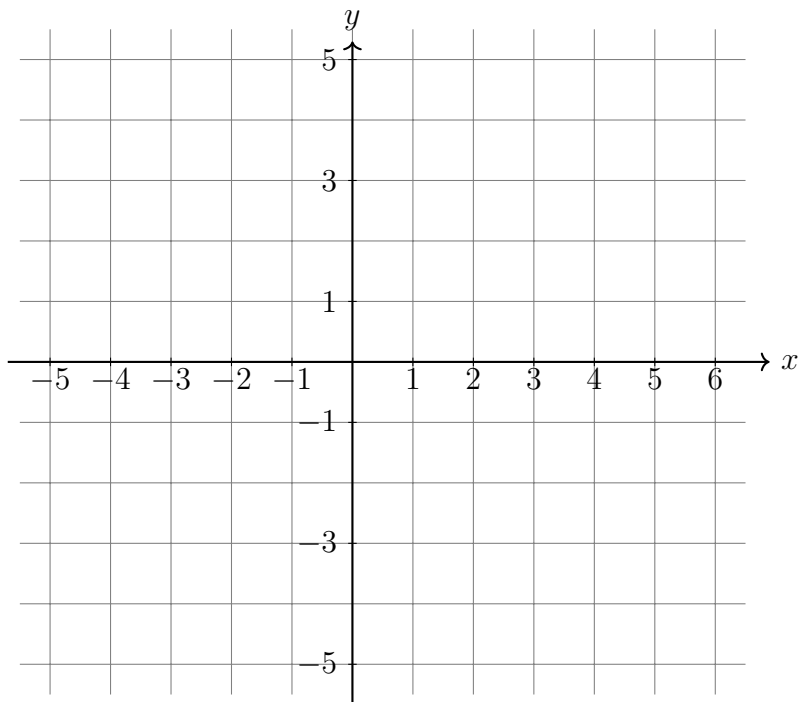
(b) Show that  $h(x) = x + \frac{4}{x^2}$ .

(c) Use part (b) to explain why the graph of  $h$  approaches the line  $y = x$  as  $|x|$  grows large. This line is called the *oblique asymptote* of  $h$ .

(d) Complete the table of values for  $h$ .

|        |    |    |    |      |     |   |   |   |
|--------|----|----|----|------|-----|---|---|---|
| $x$    | -3 | -2 | -1 | -0.5 | 0.5 | 1 | 2 | 4 |
| $h(x)$ |    |    |    |      |     |   |   |   |

(e) Sketch the graph on the axes below. Show both the vertical asymptote and the oblique asymptote  $y = x$  as dashed lines.



## Problemi della Maturità

(Use separate paper for these problems.)

### 6. PROBLEMA 2 — Sessione ordinaria 2023.

Fissato un parametro reale  $a$ , con  $a \neq 0$ , si consideri la funzione  $f_a$  così definita:

$$f_a(x) = \frac{x^2 - ax}{x^2 - a}$$

il cui grafico sarà indicato con  $\Omega_a$ .

- (a) Al variare del parametro  $a$ , determinare il dominio di  $f_a$ , studiarne le eventuali discontinuità e scrivere le equazioni di tutti i suoi asintoti.
- (b) Mostrare che, per  $a \neq 1$ , tutti i grafici  $\Omega_a$  intersecano il proprio asintoto orizzontale in uno stesso punto e condividono la stessa retta tangente nell'origine.

### 7. PROBLEMA 1 — Sessione ordinaria 2024.

Si consideri  $f_{a,b}(x) = \frac{ax^3 + b}{x^2}$ , con  $a, b \in \mathbb{R}$ .

- (a) Determinare i valori dei parametri in modo che la retta  $t$ , di equazione  $7x + y - 12 = 0$ , sia tangente al grafico di  $f_{a,b}(x)$  nel suo punto  $P$  di ascissa  $x = 1$ .

Si ponga, d'ora in avanti,  $a = 1$  e  $b = 4$ .

- (b) Studiare la funzione  $f(x) = \frac{x^3 + 4}{x^2}$  e tracciarne il grafico  $\gamma$ .